

Wick's Theorem, the Sign Problem, and Projective QMC

Emilie Huffman



ALF 2.0 Documentation, arXiv:2012.11914
F.F. Assaad and H.G. Evertz, World-line and
determinantal quantum Monte Carlo methods for spins,
phonons and electrons

Review (Partition Function, Measurements)

Wick's Theorem

Sign Problem

Projective QMC

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Partition Function (Review)

$$\hat{H} = \hat{T} + \hat{V}, \quad \hat{T} = \sum_k \hat{c}^\dagger \mathbf{T}^{(k)} \hat{c}, \quad \hat{V} = \sum_k U_k \left(\hat{c}^\dagger \mathbf{v}^{(k)} \hat{c} + \alpha_k \right)^2$$

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$$\begin{aligned} Z &= \text{Tr} \left(e^{-\beta \hat{H}} \right) \\ &= \sum_{\{l\}} \left[\prod_{k, \tau} \gamma(l_{k, \tau}) e^{\sqrt{-\Delta \tau} U_k \Phi(l_{k, \tau}) \alpha_k} \right] \det(1 + B_{L_\tau} \dots B_3 B_2 B_1), \end{aligned}$$

where

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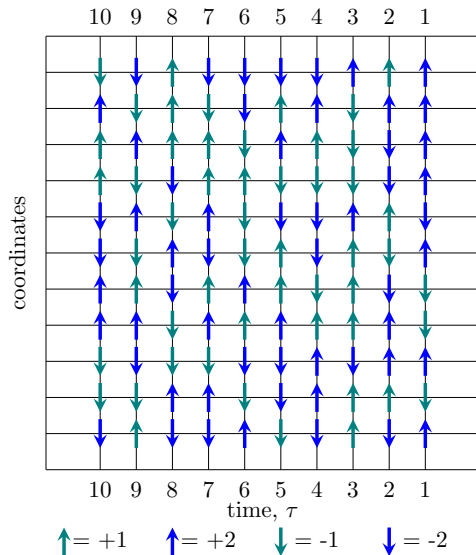
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- Configuration space consists of $\{l\} = \{l_1, l_2, \dots, l_{N_\tau \times N_k}\}$, where $l_i \in \{\pm 1, \pm 2\}$. Updates are for these Ising spin degrees of freedom.

Configuration Example (Review)



Measurements (Review)

$$\begin{aligned}\langle \hat{O} \rangle &= \frac{\sum_{\{l\}} \text{Tr} \left(e^{-\beta \hat{H}(\{l\})} \hat{O} \right)}{\sum_{\{l'\}} \text{Tr} \left(e^{-\beta \hat{H}(\{l'\})} \right)} = \frac{\sum_{\{l\}} \frac{\text{Tr} \left(e^{-\beta \hat{H}(\{l\})} \hat{O} \right)}{\text{Tr} \left(e^{-\beta \hat{H}(\{l\})} \right)} \text{Tr} \left(e^{-\beta \hat{H}(\{l\})} \right)}{\sum_{\{l'\}} \text{Tr} \left(e^{-\beta \hat{H}(\{l'\})} \right)} \\ &= \sum_{\{l\}} \langle \hat{O} \rangle_{\{l\}} \frac{\text{Tr} \left(e^{-\beta \hat{H}(\{l\})} \right)}{\sum_{\{l'\}} \text{Tr} \left(e^{-\beta \hat{H}(\{l'\})} \right)}.\end{aligned}$$

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- So we need to select $\{l\}$ configurations according to probability $p = \text{Tr} \left(e^{-\beta \hat{H}(\{l\})} \right) / \sum_{\{l'\}} \text{Tr} \left(e^{-\beta \hat{H}(\{l'\})} \right)$.

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$$\langle \hat{O} \rangle \approx \frac{1}{n} \sum_{\{l^*\}} \langle \hat{O} \rangle_{\{l^*\}}.$$

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- ▶ $\{l^*\}$ are equilibrated configurations, so are randomly selected according to the probability p .

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Measurements

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Assumptions

$$\hat{O} = T_{\tau} \hat{c}_{i_1'}^{\dagger}(\tau_1') \hat{c}_{i_1}(\tau_1) \dots \hat{c}_{i_n'}^{\dagger}(\tau_n') \hat{c}_{i_n}(\tau_n), \quad T_{\tau} : \text{time ordering}$$

$$\hat{c}_i^{\dagger}(\tau) = \left[\hat{U}_{\Phi}(\tau, 0) \right]^{-1} \hat{c}_i^{\dagger} \hat{U}_{\Phi}(\tau, 0)$$

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$$\left\langle T_{\tau} \hat{c}_i^{\dagger}(\tau_1) \hat{c}_j(\tau_2) \right\rangle_{\{l\}} = \begin{cases} \left\langle \hat{c}_i^{\dagger}(\tau_1) \hat{c}_j(\tau_2) \right\rangle_{\{l\}}, & \text{if } \tau_1 > \tau_2 \\ - \left\langle \hat{c}_j(\tau_2) \hat{c}_i^{\dagger}(\tau_1) \right\rangle_{\{l\}}, & \text{if } \tau_1 \leq \tau_2 \end{cases}$$

- Observable form:

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- Path integral:

$$\frac{\text{Tr} \left(e^{-\beta \hat{H}} \hat{O} \right)}{\text{Tr} \left(e^{-\beta \hat{H}} \right)} = \frac{\int \prod_{\alpha} d\bar{\xi}_{\alpha} d\xi_{\alpha} e^{-\sum \bar{\xi}_{\alpha} M_{\alpha\beta} \xi_{\beta}} \bar{\xi}_{i'_1, \tau'_1} \xi_{i_1, \tau_1} \dots \bar{\xi}_{i'_n, \tau'_n} \xi_{i_n, \tau_n}}{\int \prod_{\alpha} d\bar{\xi}_{\alpha} d\xi_{\alpha} e^{-\sum \bar{\xi}_{\alpha} M_{\alpha\beta} \xi_{\beta}}}$$

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- ▶ The $M_{(i, \tau), (i', \tau')}^{-1}$ quantity is called the **Green's function**.

Equal-time Green's Function

- ▶ Since we know the matrix M explicitly, we can find a form for M^{-1} . From what we derived before, we then know that the equal-time Green's function is given by

$$\begin{aligned} M_{(i,\tau),(i',\tau)}^{-1} &= - \left\langle T_{\tau} \hat{c}_{i'}^{\dagger}(\tau) \hat{c}_i(\tau) \right\rangle \\ &= (1 + B_{\tau} \dots B_1 B_{L_{\tau}} \dots B_{\tau+1})_{ii'}^{-1} = G_{\Phi}(\tau)_{ii'}. \end{aligned}$$

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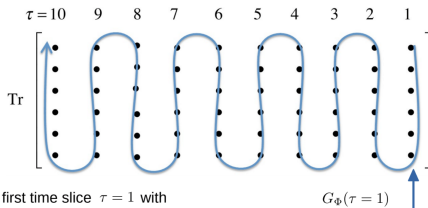
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- ▶ The equal-time Green's function is the central quantity in the algorithm.

Sweep through imaginary time



Start on first time slice $\tau = 1$ with

$$G_{\Phi}(\tau = 1)$$

Scan through the real space lattice and flip HS fields sequentially, accept the move with probability:

$$\frac{e^{-S_0(\Phi')}}{e^{-S_0(\Phi)}} \det[1 + \Delta(1 - G_{\Phi}(\tau))]$$

If move accepted, upgrade the Green function.

$$G_{\Phi}(\tau = 1) \rightarrow G_{\Phi'}(\tau = 1)$$

Outline

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Sign Problem

Projective QMC

Studying Quantum Lattice Systems Numerically

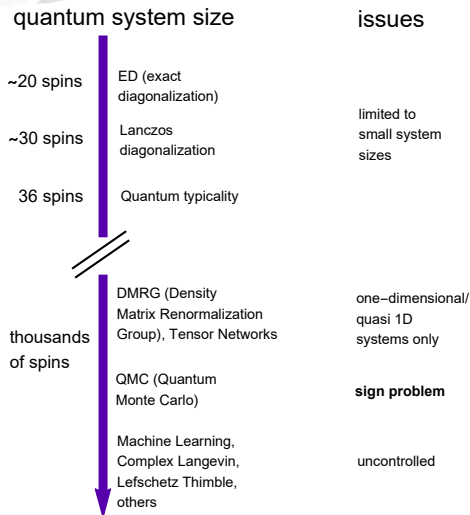


Figure: From *Dynamical Quantum Typicality*, Robin Steinigeweg.

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- ▶ Quantum mechanics involves complex numbers, and anticommutation for fermions introduces negative signs.
- ▶ ALF method of rewriting any quantum problem in terms of positive weights: $(\text{sign}(\Omega) = \text{Re}(\Omega) / |\text{Re}(\Omega)|)$

$$\begin{aligned} \langle \hat{O} \rangle_P &= \frac{\sum_C O(C) \Omega(C)}{\sum_{C'} \Omega(C')} = \frac{\sum_C O(C) \frac{\Omega(C)}{\text{Re}[\Omega(C)]} \text{Re}[\Omega(C)]}{\sum_{C'} \text{Re}[\Omega(C')]} \\ &= \frac{\sum_C O(C) \frac{\Omega(C)}{\text{Re}[\Omega(C)]} \text{sign}(\Omega(C)) |\text{Re}[\Omega(C)]| / \sum_{C''} |\text{Re}[\Omega(C'')]|}{\sum_{C'} \text{sign}(\Omega(C')) |\text{Re}[\Omega(C')]| / \sum_{C''} |\text{Re}[\Omega(C'')]|} \\ &= \frac{\left\langle \hat{O} \frac{\Omega}{\text{Re}(\Omega)} \text{sign} \right\rangle_{\bar{P}}}{\langle \text{sign} \rangle_{\bar{P}}} \end{aligned}$$

Sign Problem

- ▶ ALF samples using $\bar{P}$, and returns the two observables:

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- ▶ When $\langle \text{sign} \rangle_{\bar{P}} < 1$ the calculation time scales **exponentially** with system size.
- ▶ This is the **sign problem**.

Sign-problem-free AFQMC

- ▶ Let $\hat{\mathcal{T}}$ be an anti-unitary operator:

$$\hat{\mathcal{T}} = \mathcal{K} \hat{U}, \quad \hat{\mathcal{T}}^2 = -1, \quad \hat{\mathcal{T}} |0\rangle = |0\rangle.$$

with $\mathcal{K}$ complex conjugation, $\hat{U}$ a unitary matrix, and $[\hat{\mathcal{T}}, \hat{T}] = 0$ and $[\hat{\mathcal{T}}, \hat{v}(\Phi)] = 0$.

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(eigenvalues in complex conjugate pairs.)

Wu, Zhang, PRB 71, 155115 (2005)

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$$\hat{\mathcal{T}} \begin{pmatrix} \hat{c}_{i,\uparrow}^\dagger \\ \hat{c}_{i,\downarrow}^\dagger \end{pmatrix} \hat{\mathcal{T}}^{-1} = \begin{pmatrix} -\hat{c}_{i,\downarrow}^\dagger \\ \hat{c}_{i,\uparrow}^\dagger \end{pmatrix}$$

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Determinantal:

Wu, Zhang, PRB 71, 155115 (2005)

Huffman, Chandrasekharan, PRB 89, 111101 (2014)

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Chandrasekharan, PRD 86, 021701 (2012)

Huffman, Chandrasekharan, PRE 94, 043311 (2016)

Outline

Review (Partition Function, Measurements)

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Projective QMC

- ▶ Another way to get zero-temperature physics.

$$\begin{aligned}\langle \hat{O} \rangle_{\beta \rightarrow \infty} &= \frac{\langle \Psi_0 | \hat{O} | \Psi_0 \rangle}{\langle \Psi_0 | \Psi_0 \rangle} \\ &= \lim_{\theta \rightarrow \infty} \frac{\langle \Psi_{T,L} | e^{-\theta \hat{H}} e^{-(\beta-\tau) \hat{H}} \hat{O} e^{-\tau \hat{H}} e^{-\theta \hat{H}} | \Psi_{T,R} \rangle}{\langle \Psi_{T,L} | e^{-(2\theta+\beta) \hat{H}} | \Psi_{T,R} \rangle}\end{aligned}$$

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- ▶ Assumption:

$$\langle \Psi_{T,L/R} | \Psi_0 \rangle \neq 0.$$

- ▶ Another way to get zero-temperature physics.

$$\begin{aligned}\langle \hat{O} \rangle_{\beta \rightarrow \infty} &= \frac{\langle \Psi_0 | \hat{O} | \Psi_0 \rangle}{\langle \Psi_0 | \Psi_0 \rangle} \\ &= \lim_{\theta \rightarrow \infty} \frac{\langle \Psi_{T,L} | e^{-\theta \hat{H}} e^{-(\beta-\tau) \hat{H}} \hat{O} e^{-\tau \hat{H}} e^{-\theta \hat{H}} | \Psi_{T,R} \rangle}{\langle \Psi_{T,L} | e^{-(2\theta+\beta) \hat{H}} | \Psi_{T,R} \rangle}\end{aligned}$$

- ▶ Assumption:

$$\langle \Psi_{T,L/R} | \Psi_0 \rangle \neq 0.$$

- ▶ Trial wavefunctions have to be Slater determinants—correspond to unique ground state of a noninteracting problem:

$$\hat{H}_T^{L/R} | \Psi_{T,L/R} \rangle = E_{T,L/R} | \Psi_{T,L/R} \rangle.$$

Projective Example

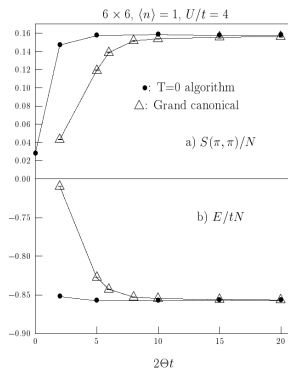


Fig. 15. Fourier transform of the spin-spin correlation functions at $\mathbf{Q} = (\pi, \pi)$ (a) and energy (b) for the half-filled Hubbard model (95). $\bullet$: PQMC algorithm. $\triangle$: FTQMC algorithm at $\beta = 2\Theta$.

Trial Wavefunction and Green's Function

Trial Wavefunction:

$$\hat{T} = \hat{c}^\dagger T \hat{c} = \hat{c}^\dagger \hat{U} D \hat{U}^\dagger \hat{c}$$

$$|\Psi_T\rangle = \prod_{n=1}^{N_{\text{Part}}} \hat{\gamma}_n^\dagger |0\rangle = \prod_{n=1}^{N_{\text{Part}}} \left(\sum_{x=1}^{N_{\text{dim}}} \hat{c}_x^\dagger U_{x,n} \right) |0\rangle = \prod_{n=1}^{N_{\text{Part}}} \left(\sum_{x=1}^{N_{\text{dim}}} \hat{c}^\dagger P \right) |0\rangle$$

Green's Function

$$G_\Phi(\tau) = \mathbb{1} - B_\tau \dots B_1 P_R \left(P_L^\dagger B_{L_T} \dots B_{\tau+1} B_\tau \dots B_1 P_R \right)^{-1} P_L^\dagger B_{L_T} \dots B_{\tau+1},$$
$$G_\Phi(\tau)^2 = G_\Phi(\tau)$$